Topology Optimization of Truss Structures to Redesign Solid Components

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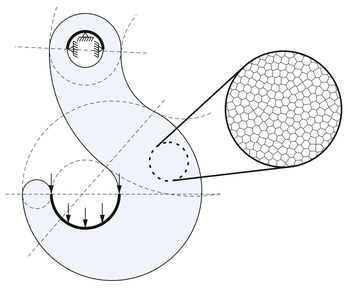
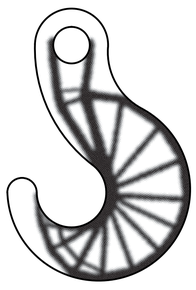
1 Motivation

Let's take a plane hook with a domain discretized by polygonal finite elements [5] →



And perform a topology optimization of the solid model (SIMP method) →



Note how the optimized structure appears more like a truss rather than a solid distribution of material? →

That's why this paper deals with the discretization and optimization of domains filled by truss elements

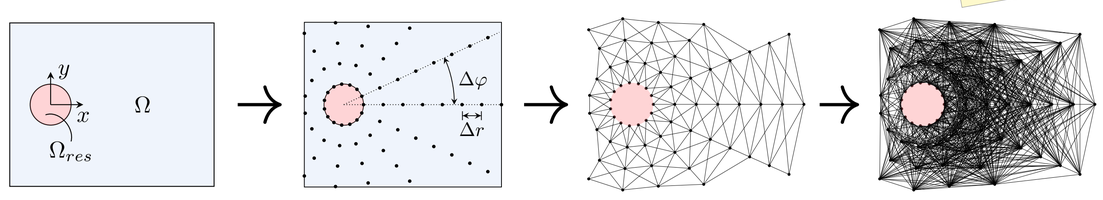
2 Finite Element Model

- Distribute a set of nodes $\mathcal{N}$ in the design domain Ω along Cartesian/polar axes
- Similar to [6], collect adjacent nodes in a connectivity matrix $\mathbf{C}^0$ to set up a base mesh and obtain structures of higher connectivity by

$$\mathbf{C}^{n_{lvl}} = \mathbf{C}^{n_{lvl}-1}\mathbf{C}^{n_{lvl}-1}$$

- The set of bars $\mathcal{B}$ is split into a physical (load carrying) set $\mathcal{B}_{phys}$ and a fictitious set $\mathcal{B}_{fict}$ containing all redundant elements

Read more about this fictitious domain approach e.g. in [4]



Domain geometry → Nodes $|\mathcal{N}^{(0)}| = 66$ → Base mesh $n_{lvl} = 0$ → Ground Structure $n_{lvl} = 2$

3 Optimization Algorithm

Weight optimization

$$\min_{\mathbf{a}} \; V = \sum l_i a_i \quad i \in \mathcal{B}_{phys}$$

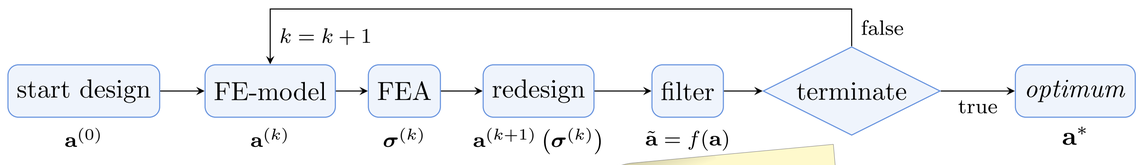
$$\text{s.t.} \quad a^l < a_i \leq a^u$$
$$|\sigma_i| = \sigma^u$$

Aim for homogeneous stress distribution at the max. permissible amount

Redesign formula based on [1]

- Vary cross-sectional areas $\mathbf{a}$ based on the element's stress capacity
- Penalization of redundant elements with μ

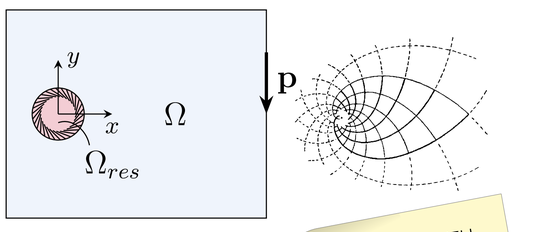
$$a_i^{(k+1)} = \begin{cases} \min\left(a_i^{(k)}\frac{\sigma_i^{(k)}}{\sigma^u},\; a^u\right) & , a_i^{(k+1)} > a^l \\ \mu = 10^{-12} & , a_i^{(k+1)} \leq a^l \end{cases} \quad i \in \mathcal{B}$$



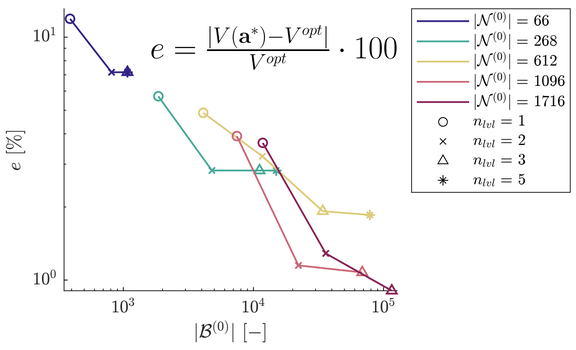
The Heaviside function is used for filtering to improve contouring and convergence [2]

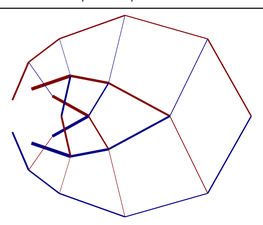
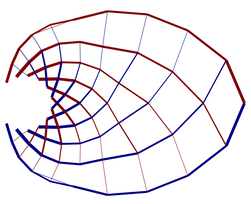
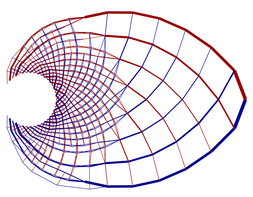
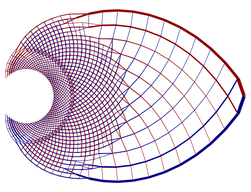
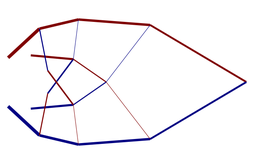
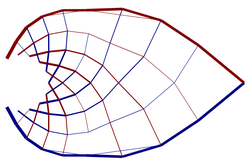
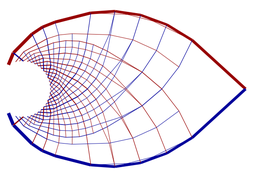
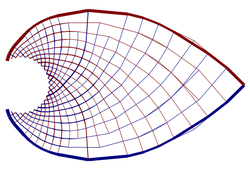
4 Verification

The optimality is evaluated based on MICHELL's optimal solution V^{opt} [3]



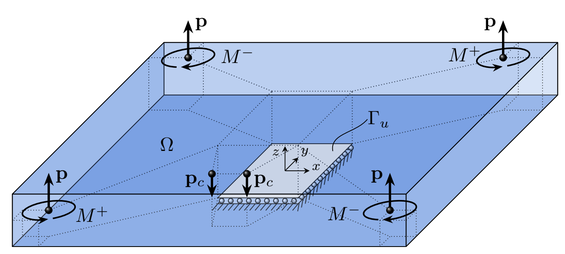
The original MICHELL structure [3] for visual comparison

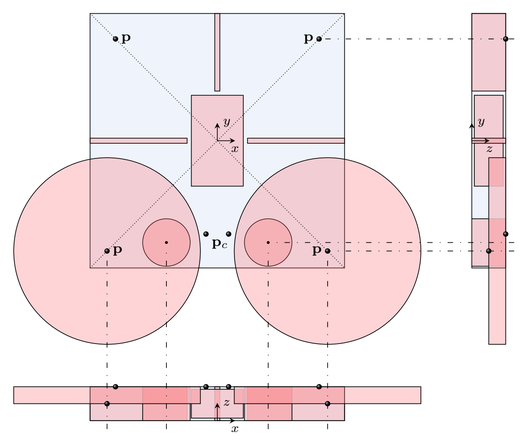


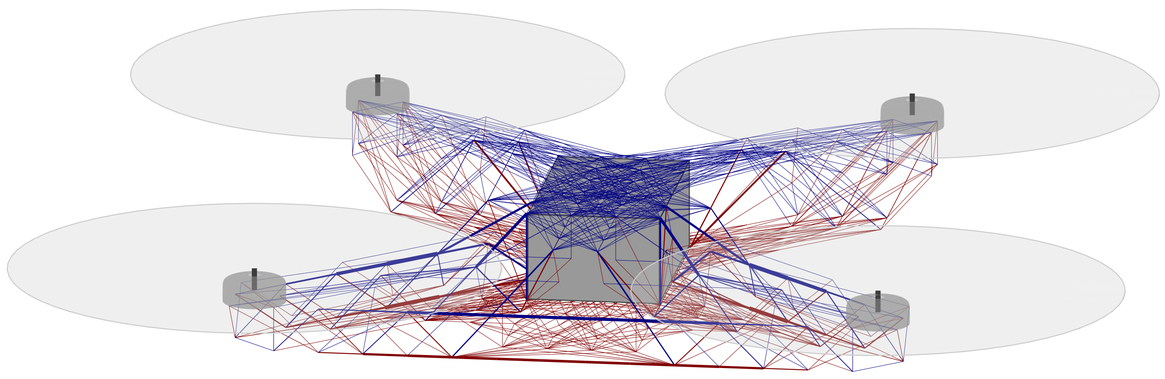
n_{lvl}	$\lvert\mathcal{N}^{(0)}\rvert = 66$	$\lvert\mathcal{N}^{(0)}\rvert = 268$	$\lvert\mathcal{N}^{(0)}\rvert = 612$	$\lvert\mathcal{N}^{(0)}\rvert = 1716$
1				
3				

5 Demonstration

- Model of a drone frame
- Test on behavior with more complex geometry and boundary conditions







6 Conclusion

- Algorithm converges to MICHELL's optimal solution
- High computational effort for high optimality (up to several hours run time)
- Nevertheless errors $< 1\%$ obtained
- Contouring issues for densely filled ground structures
- Stress field-based pre-orientation of the ground structure could be helpful

References

[1] BAIER, H. ; SEESSELBERG, C. ; SPECHT, B. : *Optimierung in der Strukturmechanik*. Springer-Verlag, 1994

[2] GROEN, J. P. ; SIGMUND, O. : Homogenization-based topology optimization for high-resolution manufacturable microstructures. In: *International Journal for Numerical Methods in Engineering* 113 (2018), Nr. 8, S. 1148–1163

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[6] ZEGARD, T. ; PAULINO, G. H.: GRAND - Ground structure based topology optimization for arbitrary 2D domains using MATLAB. In: *Structural and Multidisciplinary Optimization* 50 (2014), Nr. 5, S. 861–882